A Study on the Generalization of Sarcasm Detection Models

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Abstract

Sarcasm is the use of words which mean the opposite of what you really wish to say, with the intention of conveying a point, being funny, or to show irritation. Sarcasm detection in natural language processing remains a challenge due to the ambiguity and context-dependence of sarcastic expressions. Existing models often perform well on the domain they are trained on but struggle with unseen domains. This research investigates whether training a sarcasm detection model on a combined dataset encompassing multiple text domains (News Headlines, Twitter, Reddit) improves generalizability compared to models trained on individual domains. We implemented three machine learning models (SVM, Bi-LSTM, BERT) and evaluated their performance on individual datasets. Our results confirm that models trained on a specific domain perform poorly on unseen domains. We improved prediction performance for multiple text domains by combining data from various sources (average of accuracy on three dataset: 0.732). However, in this process, we found that our performance fell short of the predictive performance of a model learned by focusing on a specific dataset.

1. Introduction

Sarcasm is usually used to express strong opinion and feeling, but it involves saying the opposite of what one truly means, often in a tone that contrasts with the literal meaning of the words. Because of these characteristics of sarcasm, detecting sarcasm using only text without prior information about voice tone, facial expression, or situation remains a challenging task. However, sarcasm detection is a task that is a foundation of various applications of natural language processing (NLP) in fields such as marketing or sentimental analysis. Because of this importance, research on sarcasm detection has been actively conducted. In early studies for sarcasm detection, data was mainly used from Reddit or Twitter, since they have specific annotations indicative of sarcastic comments. However, due to the ambiguity of sarcasm, other users not agreeing with the author's annotation, and the incorrect use of annotations, Reddit data and Twitter data have a lot of language noise. Diverse studies have been conducted to solve these problems. In order to take into account incorrectly used annotations in the process of collecting reddit data, Khodak et al. removed the texts of users who were using sarcasm annotation for the first time (Khodak et al., 2018). Misra et al. created a sarcasm detection model using News Headline data (Misra, n.d.). News headlines, unlike Reddit or Twitter data, have no typos and have a formal structure, making model learning easier. Wang et al. used various types of data such as images in addition to text data to create a sarcasm detection model for Twitter data (Wang et al., 2022). Through the process of reviewing existing studies, we were able to confirm that even when algorithms and pipelines are used, learning and verification are conducted separately depending on the data source. However, fields requiring sarcasm detection deal with various texts generated from sources outside of Headlines, Twitter and Reddit. For this reason, we thought that building a sarcasm detection model that can be universally applied even to texts with unknown sources would be a great contribution to the field. Unlike existing studies, we focused on improving average performance on various datasets instead of improving model performance on specific datasets.

2. Methods

2.1. Data Collection

We collected data from three different sources widely used in sarcasm detection research. Each entry in these datasets contains text and a binary label that indicates whether the text is sarcastic or not. 1 signifies that sarcasm was detected, and 0 signifies that no sarcasm was detected.

News Headlines dataset: News headlines data for sarcasm detection was provided by FlorianKibler sarcasm_dataset_en dataset from Hugging Face (Floriankibler, n.d.) (Misra, n.d.). This dataset contains news headlines from TheOnion and HuffPost. TheOnion publishes satirical articles on current events. Therefore, the news headlines from TheOnion were considered sarcastic data and the news headlines from HuffPost were assigned non-sarcastic labels.

Twitter dataset: Twitter data was obtained from pre-

vious research for sarcasm detection [twitter dataset]. This dataset contains English tweets and its label is assigned based on the hashtag (e.g., #sarcasm).

Reddit dataset: Reddit data was collected from a previous study (Floriankibler, n.d.). Labels of comments were assigned based on users' conventional annotation for sarcasm (e.g., "/s").

2.2. Data Preprocessing and Sampling

We converted all text to lower-case and removed punctuation from the text. For this research, we only used texts with 3 to 30 words inclusive to handle noise considering the distribution of texts (See Results). To check the performance of the sarcasm detection model learned from a specific dataset, we sampled the same number of data from each dataset. Then, we split the data into training, validation, and test set with a 6:2:2 ratio. In this process, we kept the label ratio at 1:1.

Additionally, we made combined datasets by sampling with various ratios from each dataset to build a general sarcasm detection model. Table 1 shows examples of sampling and splitting data. The (1:1:1) represents the ratio of sampled data from the Headlines, Twitter and Reddit datasets, respectively.

2.3. Models

To build sarcasm detection models, we implemented three machine learning models widely used for text classification: Support Vector Machine (SVM), Bidirectional LSTM (Bi-LSTM), and Bidirectional Encoder Representations from Transformers (BERT). We used existing libraries to implement those models including Scikit-learn, PyTorch, and Transformers. To vectorize input texts, we used the TF-IDF method for SVM and GloVe word embedding for Bi-LSTM. Since we used a pretrained BERT model, "bert-base-uncases", raw texts were used as input for the BERT model.

2.4. Model Evaluation

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1_Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

3. Results

3.1. Data Distribution

Fig 1. shows the data distribution according to the length of texts for each datasets after preprocessing. In this project, we focused on building a sarcasm detection model only using given text without additional information. However, the Twitter and Reddit datasets contain text with only one word or number since comments on the posts are also included in the datasets. In addition, they also have data consisting of repetitions of single words and it corresponds to extremely long sentences. To reduce this noise, we only used texts with length between 3 and 30 inclusive. Through this process, we were also able to prevent the input data from being filled with too many zeros by zero-padding.

3.2. Baseline

We trained SVM, Bi-LSTM, and BERT models using a single data set. We used 5 epochs for Bi-LSTM and BERT, and chose the model with the highest validation accuracy. We repeated this process for datasets of 1500, 3000, and 4500 samples. As you can see in Figure 2, the performance of the model depends more on the model than on the size of the dataset. Although BERT took the longest time to train, it performed better than the other models on every dataset.

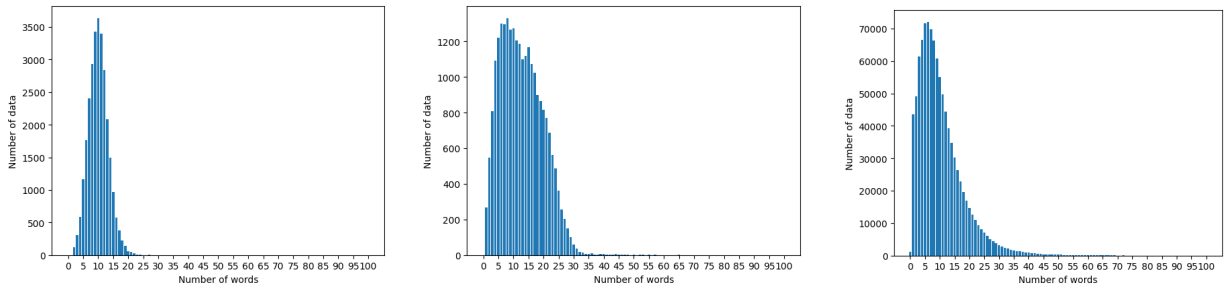
Additionally, we were able to confirm that the performance of the classifier differed significantly depending on the dataset. For all combinations of models and dataset sizes, better prediction performance was achieved in the following order: News headlines, Twitter, and Reddit datasets. Unlike the news headlines written by professionals in a formal manner, Twitter and Reddit data have more language noise (Misra, n.d.). In particular, the classifier trained on Reddit data had a validation accuracy of less than 0.7. Unlike the annotations of Twitter, the annotations of Reddit are frequently misused (Khodak et al., 2018). Furthermore, we attempted to make predictions on other datasets using a classifier learned from a specific dataset, then measured accuracy and used this as a baseline [Table 2].

Unlike the news headlines written by professionals in a formal manner, Twitter and Reddit data have more language noise (Khodak et al., 2018). In particular, models trained using the news headline dataset and Reddit dataset showed the worst prediction performance on each other's test sets. Even though Twitter and Reddit are common social media data sets, they did not show good prediction performance (accuracy under 0.6). We suspect that this is because anno-

Table 1. Example of Data Sampling and Splitting

TRAIN SOURCE	TRAINING	VALIDATION	TEST
HEADLINES	HL: 2700	HL:900	HL:900
COMBINED DATA (1:1:1)	HL: 900 TW: 900 RD: 900	HL: 300 TW: 300 RD: 300	X

HL: NEWS HEADLINES, TW: TWITTER, RD: REDDIT



(a) Headline (b) Twitter (c) Reddit

Figure 1. Text Length Distribution for Each Dataset

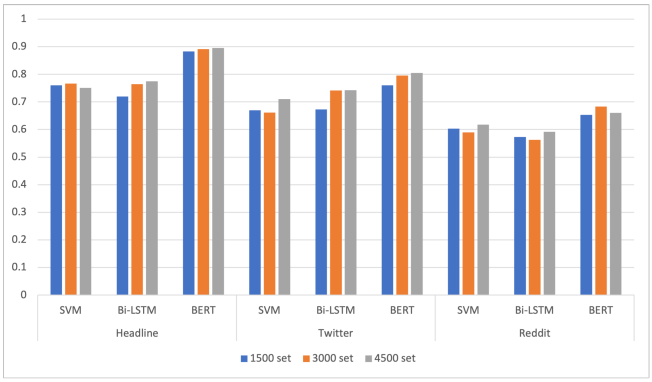


Figure 2. Model Accuracy on Validation set

tations of Reddit are more noisy compared to annotations of Twitter even though they are both data freely written by users (Khodak et al., 2018)

3.3. Sarcasm Detection Model Learned with Combined Data

In order to make a general sarcasm detection model that can be used on various data sources, we conducted training using a combined dataset sampled from each data source while maintaining . In this process, we sampled data at various ratios to find the optimal combination, made predictions on the test set used in the baseline, and evaluated performance with accuracy, precision, recall, and F1 score. To obtain the robust results, we duplicated these processes 5 times for each ratio.

Table 2. Model performance on test sets for Datasets

TRAIN SOURCE	HEADLINES	TWITTER	REDDIT
HEADLINE	0.91	0.554	0.537
TWITTER	0.59	0.83	0.594
REDDIT	0.427	0.545	0.66

Table 3. Average performance

RATIO	ACCURACY	PRECISION	RECALL	F1
1:1:1	0.718	0.716	0.730	0.722
1:3:1	0.721	0.739	0.686	0.710
2:2:1	0.723	0.743	0.680	0.709
3:1:1	0.728	0.733	0.714	0.723
3:1:2	0.721	0.728	0.706	0.716
2:3:1	0.732	0.745	0.708	0.724
3:2:1	0.726	0.746	0.678	0.709

Table 3 presents the performance of classifiers trained on data sampled at various ratios. The split ratio that showed the best performance was Headline 2: Twitter 3: Reddit 1. Table 4 shows performance of the model learned with this combined set for each dataset.

This result suggests that while it didn't achieve accuracy comparable to that on the training and test data from the same source, it showed superior generalization ability com-

Table 4. Performance of Best Ratio Model

	HEADLINE	TWITTER	REDDIT
ACCURACY	0.835	0.768	0.591
PRECISION	0.859	0.778	0.597
RECALL	0.803	0.750	0.570
F1	0.830	0.763	0.581

pared to a classifier trained on a specific dataset.

4. Process

We applied the Ensemble Method using the Bert model trained using the combined dataset.
1) 1500 combined dataset, 3000 combined dataset, 4500 combined dataset
Accuracy: (Headline: 0.858 Twitter: 0.724 0.632)
2) 5 models that used 1500 combined dataset
Accuracy: (Headline: 0.832 Twitter: 0.728 0.601)
3) a model that used datasets of all ratios mentioned in the results section Accuracy: (Headline: 0.874 Twitter: 0.793 0.613)

Compared to the best ratio model (Table 4), the third ensemble model shows slight improvement. However, considering that the training time increased sevenfold, we do not see this as a meaningful performance improvement.

We randomly sampled the data, and since Twitter and Reddit data are noisy, we expected performance to improve through the Ensemble Method. However, contrary to expectations, there was no significant difference from using a single combined dataset. We decided that this was because our sample size was sufficient to reflect the characteristics of the dataset.

Although our goal was not to improve performance on a specific dataset, we hoped to improve performance on Reddit data as we experimented. However, in order to improve performance on Reddit data, it seems that priority should be given to finding a way to preprocess Reddit text rather than using data without noise.

5. Contributions

5.1. Datasets

Our work wouldn't have been possible without referencing existing research and resources conducted on sarcasm detection methods. (Cai et al., 2019) produced a multi-modal sarcasm detection model and an accompanying dataset that we used to train our models. Reddit data was sourced from a dataset compiled by user FlorianKibler on HuggingFace (Floriankibler, n.d.), as well as the Self-Annotated Reddit Corpus created by (Khodak et al., 2018). News headlines and work done by (Jamil et al., 2021) were also essential in the development and training of our models.

5.2. Open Source Libraries, Tools, and Code

For dataset manipulation and model development and access we used the HuggingFace Datasets and Transformers libraries respectively, and for data processing we relied on popular libraries such as Pandas and NumPy. Data visualization was facilitated by the matplotlib and seaborn libraries, which enabled us to analyze and communicate our results. Finally, we utilized scikit-learn, NLTK, and PyTorch for our machine learning tasks. For handling the Twitter data, we found open source code examples Eugene Siow's practical-ml repository (Eugene, n.d.) which proved extremely helpful, specifically the notebook on implementing RoBERTa for sarcasm detection. We used Google Colab for our programming environment.

6. Project Code

The code to recreate our experiments is available at https://drive.google.com/drive/folders/1MQ-v815whEkJRvA3hEpLFFGyRbTCD-VM?usp=sharing. You must be logged into a usc.edu email to access the files.

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